Configure network devices, such as hubs and switches within a network topology using Packet Tracer software

Aim

To create and configure a network topology using a **Hub**, **Switch**, and **PCs** in Cisco Packet Tracer and verify communication using the **Ping** command.

Software Required

- Cisco Packet Tracer

Devices Required

- 1 Hub
- 1 Switch
- 4 PCs
- Copper Straight-Through cables

Part A: Create a Network Using a Hub

Step 1: Open Cisco Packet Tracer

1. Open **Cisco Packet Tracer**.
2. Select **File → New** to create a new project.

Step 2: Add PCs

1. At the bottom-left corner, click **End Devices**.
2. Select **PC-PT**.
3. Drag and place **4 PCs** in the workspace.
4. The devices will appear as:
    - PC0
    - PC1
    - PC2
    - PC3

Step 3: Add a Hub

1. Click **Network Devices**.
2. Select **Hubs**.
3. Choose **Hub-PT**.
4. Drag the hub into the workspace.

Step 4: Connect the PCs to the Hub

1. Click the **Connections** icon (lightning bolt symbol).
2. Select **Copper Straight-Through** cable.
3. Connect each PC to the Hub as follows:

Connection	Port
PC0 → Hub	PC0 FastEthernet0 → Hub Port 1
PC1 → Hub	PC1 FastEthernet0 → Hub Port 2
PC2 → Hub	PC2 FastEthernet0 → Hub Port 3
PC3 → Hub	PC3 FastEthernet0 → Hub Port 4

4. Wait for a few seconds.
5. The connection indicators should turn **green**.

Step 5: Configure IP Addresses

Configure all PCs in the same network.

For each PC:

1. Click the PC.
2. Select the **Desktop** tab.
3. Click **IP Configuration**.
4. Select **Static**.
5. Enter the IP address and subnet mask.

Use the following IP configuration:

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Leave the **Default Gateway** blank because no router is used.

Step 6: Test Connectivity

1. Click **PC0**.
2. Select **Desktop → Command Prompt**.
3. Type:

ping 192.168.1.2

4. Press **Enter**.

Expected output:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Similarly, test communication with other PCs:

ping 192.168.1.3
ping 192.168.1.4

If replies are received, the network is working correctly.

Part B: Create a Network Using a Switch

Step 1: Add a Switch

1. Click **Network Devices**.
2. Select **Switches**.
3. Choose **Cisco 2960 Switch**.
4. Drag the switch into the workspace.

Step 2: Add Four PCs

1. Click **End Devices**.
2. Select **PC-PT**.
3. Place four PCs around the switch.

Step 3: Connect PCs to the Switch

Use **Copper Straight-Through** cables.

Connection	**Port**
PC0 → Switch	FastEthernet0 → FastEthernet0/1
PC1 → Switch	FastEthernet0 → FastEthernet0/2

Connection **Port**

PC2 → Switch FastEthernet0 → FastEthernet0/3

PC3 → Switch FastEthernet0 → FastEthernet0/4

Wait until the connection lights turn **green**.

Step 4: Configure IP Addresses

Assign IP addresses in the same network.

Device	IP Address	Subnet Mask
PC0	192.168.2.1	255.255.255.0
PC1	192.168.2.2	255.255.255.0
PC2	192.168.2.3	255.255.255.0
PC3	192.168.2.4	255.255.255.0

Step 5: Test Connectivity

Open the Command Prompt on PC0:

ping 192.168.2.2

Then test:

ping 192.168.2.3
ping 192.168.2.4

Successful replies confirm that all devices are connected and communicating through the switch.

Conclusion:

The network topology was successfully created using **Hub and Switch devices in Cisco Packet Tracer**. IP addresses were assigned to all PCs, and network connectivity was verified successfully using the **Ping command**.